13 Liquid A and liquid B are mixed together in the ratio 2 : 13 by volume to make liquid C.

Liquid A has density 1.21 g/cm^3
Liquid B has density 1.02 g/cm^3

A cylindrical container is filled completely with liquid C.
The cylinder has radius 3 cm and height 25 cm.

Work out the mass of the liquid in the container.
Give your answer correct to 3 significant figures.
You must show all your working.

Cylinder volume = $\pi \times 3^2 \times 25 = 225\pi \approx 706.858$ cm^3.

A has 2/15 of the volume; B has 13/15.

Mass = $225\pi[(2/15)\times 1.21 + (13/15)\times 1.02]$

≈ 738.903 g ≈ 739 g (3 s.f.).

4 marks total
Cylinder volume
Ratio volumes
Density × volume
739 g

..... g

(Total for Question 13 is 4 marks)

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